Neoadjuvant proton beam irradiation followed by transscleral resection of uveal melanoma in 106 cases

Gregor D Willerding,[1,2] Dino Cordini,[1,3] Lutz Moser,[4,5] Lothar Krause,[6] Michael H Foerster,[1,7] Nikolaos E Bechrakis[8]

For numbered affiliations see end of article.

Correspondence to Professor Nikolaos E Bechrakis, Department of Ophthalmology, Medical University Innsbruck, Anichstr. 35, Innsbruck A-6020, Austria; nikolaos.bechrakis@i-med.ac.at

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ABSTRACT

Aims To describe results after neoadjuvant proton beam irradiation followed by transscleral resection of large uveal melanoma.
Methods Retrospective interventional case series, including 106 consecutive patients. Local tumour control, enucleation and metastasis development were assessed with survival curves. Predictors of local recurrence and metastasis were investigated with log-rank testing.
Results Mean follow-up was 3.2 years. Local recurrence occurred in five cases with an estimated risk of 4.2% and 10.4% at 3 and 5 years after treatment, respectively. Enucleation was performed in 10 cases with an estimated risk of 9.2% and 18.4% at 3 and 5 years, respectively. Significant risk factors for local recurrence were not evident. Metastasis was estimated to occur in 28.4% at 3 years and 40.3% at 5 years, correlating with patient's age only (p=0.01). Seventy four patients (69.8%) underwent vitreoretinal surgery for complications after tumour resection. Median visual acuity (VA) was 20/50 at diagnosis and 20/400 in the third year after treatment. VA preservation of 20/200 or better was achieved in 33 patients (31.1%).
Conclusion Neoadjuvant proton beam irradiation may help to prevent local recurrence after transscleral resection. Additional vitreoretinal surgery was frequently needed in the presented series. The majority of patients avoided enucleation and functional blindness.

INTRODUCTION

Radiotherapy is a successful treatment in most uveal melanomas. In large tumours, eye preservation rate and functional outcome are markedly impaired after various modes of radiotherapy due to radiogenic complications. This has been shown both for brachytherapy as well as for charged particle irradiation.[1–5]

More than 40 years ago, Stallard, Müller and later Foulds developed the technique of transscleral resection (TSR) as a primary treatment of melanomas located both anteriorly in the ciliary body as well as more posteriorly in the choroid. Adjunctive radiotherapy after TSR for control of obvious or occult residual malignant cells has been realised with plaque radiotherapy or charged particle irradiation.[6 7] Our group reported results after TSR with adjuvant ruthenium plaque irradiation in 2009.[8] In that series, a local recurrence rate of roughly 20% at 5 years was noted, and local recurrence was found to be associated with systemic metastasis. To further improve local tumour control, we started performing neoadjuvant proton beam irradiation (PBI) prior to TSR of large uveal melanoma in a prospective protocol from 2004 onwards, and present the results herein.

MATERIALS AND METHODS

The demographic and irradiation-related data were extracted from the computerised dataset of all patients treated with PBI at our centre. Further information was collected from the medical records.

All patients who underwent PBI and planned TSR in arterial hypotension between August 2004 and July 2010 were included in this analysis. Patients were eligible for this treatment protocol if they had choroidal or ciliochoroidal, non-metastasised melanoma with a minimum tumour thickness of 7 mm (in two patients tumour thickness was 6 mm) and a largest basal diameter of <23 mm.

The procedure was primarily offered to patients with a visual acuity (VA) of 20/200 or better, but in six patients, VA was <20/200. All patients expressed a high motivation to retain their eye. Apart from abdominal ultrasound and chest X-ray, a medical work-up to assess the eligibility for arterial hypotension included a neurological examination, an electroencephalogram, Doppler ultrasound of the external carotid arteries, 24 h blood pressure measurement as well as 24 h echocardiogram and a stress echocardiogram.

Informed consent was obtained before treatment. The tenets of the Declaration of Helsinki were respected. The data accumulation was carried out with approval from the local institutional review board (reference number EA4/037/13). Prior to PBI, four tantalum clips were sutured on the sclera under general anaesthesia. In 94 patients, orbital MRI was incorporated in treatment planning with OCTOPUS software.[9 10] In the remaining cases, the target volume was conventionally planned with EYEPLAN software according to ultrasound measurements of the tumour dimensions relative to the tantalum clips.[11] The safety margin for irradiation was set to 2.5 mm, that is, 50% of the prescribed dose 2.5 mm outside the defined clinical target volume. PBI was performed with four fractions, each 15 cobalt gray equivalent (54.5 Gy), over 4 days in a lid-sparing manner in all but two cases. After a mean interval of 2 weeks (minimum 4 days, maximum 6 weeks) post irradiation, the tumour resection was performed similar to a technique reported in detail recently.[12] A lamellar scleral dissection and excision of the inner scleral lamella with the uveal tumour after partial one-port pars plana vitrectomy was followed by closure of the scleral flap and instillation of a sodium hexafluoride bubble intravitreally (0.5–1.0 mL). In case of postoperative complications, secondary vitreoretinal procedures or cataract removal were performed as needed.

The average interval for the first follow-up examination after TSR was 1 month, individually ranging from several days to 6 months. Further examinations were scheduled in intervals of 3–6 months during the first 2 years, 3–12 months in the third year, 6–12 months up to 5 years after PBI and 6–24 months thereafter, depending on the need of secondary treatment. Six-monthly liver ultrasound for detection of metastasis was recommended in all patients.

Any progressive pigmented intraocular mass as well as progressive residual tumour was regarded as recurrence, and was treated with additional focally destructive measures or enucleation.

Statistics included Kaplan–Meier analysis and log-rank test for local recurrence, metastasis and eye retention. VA was analysed by box-plot analysis on regular follow-up intervals.

RESULTS

One hundred and six patients with choroidal (n=20) or ciliochoroidal (n=86) melanoma were selected for TSR in arterial hypotension following PBI between 3 August 2004 and 6 July 2010. One further case was excluded from this series because a medulloepithelioma was diagnosed histopathologically after completion of neoadjuvant PBI and TSR. Mean patient age was 57.7 years (range 25–81 years). Fifty five (51.9%) of our patients were male.

Tumour characteristics are listed in the table 1. According to the American Joint Committee on Cancer Tumor-Node-Metastasis Protocol (2009), the tumours were classified as T1 in one case, T2 in two cases, T3 in 62 cases and T4 in 41 cases.

Anterior chamber involvement and retinal detachment were present at treatment in 23.6% and 83.0%, respectively. The tumour distance to the optic disc and fovea was <5 mm in 21 and 25 patients, respectively.

Extraocular extension was found in 13 patients (12.3%); in six patients, anterior extraocular growth with a maximum extension of 3 mm was present. Another seven patients showed posterior extrascleral growth associated to the vortex veins with 5 mm width in one case and <3 mm in the remainder. These findings were included in the irradiation field in all cases.

Histopathology reported spindle cell melanoma in 52 (49.1%), epithelioid cell melanoma in 10 (9.4%) and mixed cell melanoma in 44 cases (41.5%). Cytogenetic analysis of tumour tissue was not performed at that time.

Mean follow-up after radiotherapy was 38.1 months (median 34.4 months, range 1.3–96.6 months). Ninety four (88.7%), 52 (49.1%) and 22 patients (20.8%) completed 1, 3 and 5 years of follow-up, respectively.

Table 1 Tumour characteristics at treatment

Characteristics	Mean±SD (mm)	Median (mm)	Range (mm)
Tumour thickness	10.6±1.9	10.5	5.9–14.7
LBD	16.4±2.6	16.5	7.0–21.7
TDO	9.3±4.4	9.1	0.8–21.0
TDF	9.0±4.5	8.5	0.0–20.4

LBD, largest basal diameter; TDF, tumour distance to fovea; TDO, tumour distance to optic disc.

Kaplan–Meier estimate of systemic metastasis was 28.4% and 40.3% at 3 and 5 years, respectively (figure 1A). Patient age at diagnosis, ciliary body infiltration, largest basal diameter, tumour distance to optic disc/fovea, histological subtype and presence of retinal detachment at diagnosis were evaluated as risk factors for local tumour recurrence and metastasis.

Local tumour recurrence developed in five patients (4.7%) between 2.5 and 57.9 months after TSR. The 3-year and 5-year

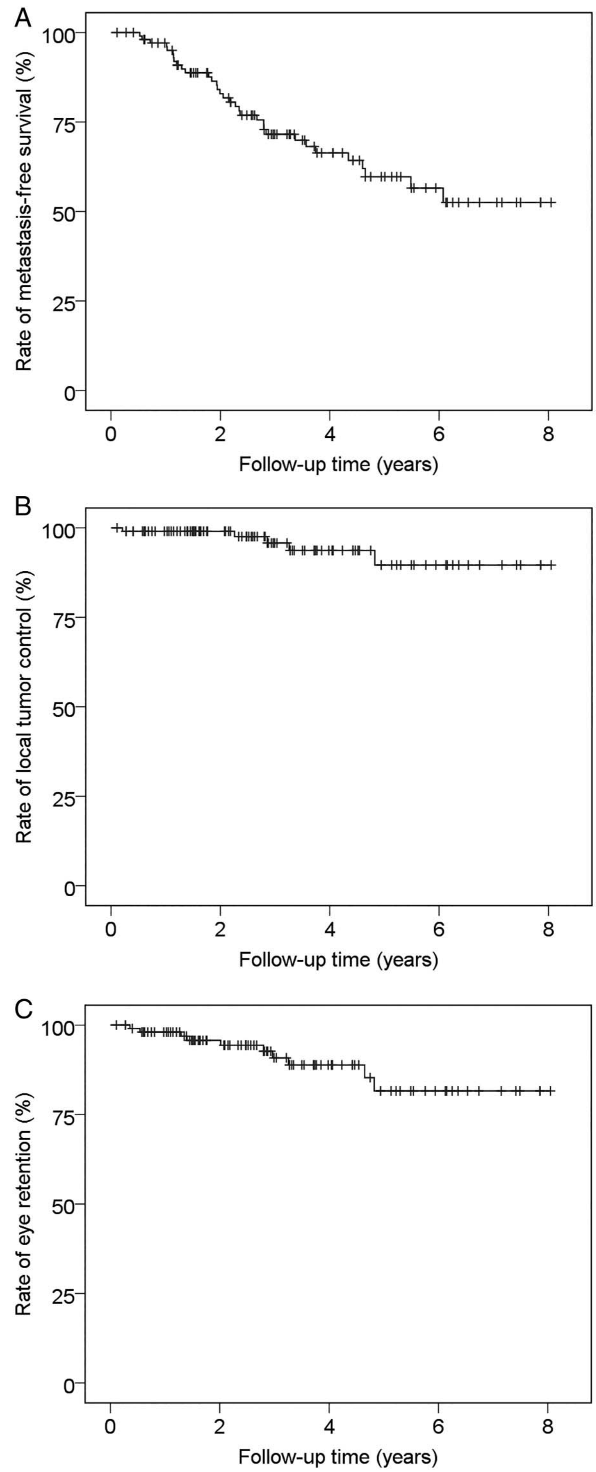

Figure 1 (A) Kaplan–Meier estimate of metastasis-free survival. (B) Kaplan–Meier estimate of local tumour control. (C) Kaplan–Meier estimate of eye retention.

Kaplan–Meier estimates for local recurrence were 4.2% and 10.4%, respectively (figure 1B). All recurrences occurred intraocularly. In three of the five recurrent cases, the initial largest tumour diameter measured <16 mm. In this respect, the largest tumour basal diameter was not significantly associated with local recurrence.

In two patients, recurrence occurred at the posterior choroidectomy margin, in one patient at the anterior excision margin and in another patient at a lateral choroidectomy margin. The treatment was enucleation in these four patients. In the fifth patient, the recurrent tumour was unifocal, but clinically non-contiguous with the primary tumour site. In this case, additional focal brachytherapy was performed with successful tumour control up to the end of follow-up 15.5 months later. No episcleral or orbital recurrence was noted during follow-up. None of the analysed parameters proved to be a significant risk factor for local tumour recurrence.

Regarding metastasis, only minimum age of 50 years at treatment proved to be a significant predictor in univariate analysis.

Residual tumour was noted or suspected during surgery or during early follow-up after clearing of intraocular haemorrhage in 70 patients (66%). In these cases, no significant progression was noted in any case during follow-up, and, therefore, these patients were classified as having residual tumours and not recurrences. Additional 'consolidation' treatment at these residual tumours was done in 10 cases at the time of vitreoretinal surgery (photocoagulation in six cases and endoresection in four cases).

Secondary enucleation was performed in 10 patients (9.4%). The reasons were tumour recurrence in four patients, retinal detachment and suspected (histologically not proven) recurrence in one patient and opaque media and patient preference in one. In four patients who initially had tumours with a maximum tumour basal diameter of 15.4, 16.6, 19.0 and 19.8 mm, enucleation was performed after loss of function and phthisis bulbi. The Kaplan–Meier estimate for vision loss was 9.2% at 3 years and 18.4% at 5 years (figure 1C).

Additional vitreoretinal surgery after TSR was necessary in 74 patients (69.8%). Indications were intraocular haemorrhage in 20 patients, retinal detachment in 13 patients and both in 40 patients. One patient received vitrectomy for a dropped nucleus at the time of cataract surgery. In the 53 patients who underwent surgery for retinal detachment, 46 (86.8%) had an attached retina at the last follow-up, 28 of whom had silicone oil tamponade in place.

Cataract surgery was performed in 94 of initially 100 phakic patients. In 66 cases, this was done during concurrent vitreoretinal surgery. Nine patients were left aphakic.

The distribution of VA postoperatively is illustrated in figure 2. Of the 94 eyes with an initial VA of 20/200 or better, 44 patients (46.8%) remained in this VA category (20/200 or better) in the first year of follow-up. In the third and the fifth year, 26 out of 61 (42.6%) and 10 out of 24 (41.7%) kept a VA of 20/200 or better, respectively. Of all patients in this series, less than counting fingers was present in 17 out of 106 (16.0%) after 1 year, 12 out of 70 (17.1%) at 3 years and 7 out of 26 (26.9%) at 5 years of follow-up. Median VA was 20/50 at diagnosis and 20/400 in the third year after treatment.

Rubeotic glaucoma developed in 17 patients (16.0%) up to the last follow-up. This group did not differ significantly in terms of tumour size, ciliary body or total dose volume. Two of the patients were diabetic, and 10 eyes had silicone tamponade in place at last follow-up.

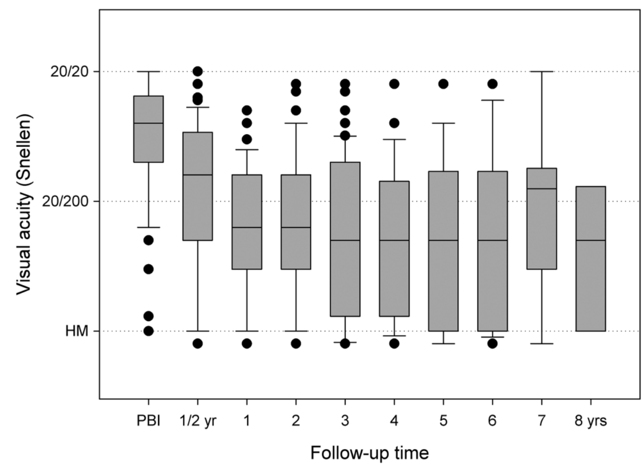

Figure 2 Distribution of postoperative visual acuity (box and whisker plots with outliers). HM, hand movements; PBI, proton beam irradiation.

DISCUSSION

Several studies indicate that adjunctive radiotherapy improves local tumour control after TSR of uveal melanoma.[6 8 13 14] In our experience, adjuvant ruthenium-106 brachytherapy with a 20 mm plaque did reduce local recurrence after TSR, but still local tumour control was felt to be less than optimal.[8] Therefore, we replaced adjuvant brachytherapy by neoadjuvant PBI in 2004 in order to achieve better local tumour control.

The patients treated with TSR and adjuvant ruthenium-106 brachytherapy in our previously published series were largely comparable in terms of demographic or tumour-related data. Our change to neoadjuvant PBI seems to be associated with a decrease in local recurrence; however, the lack of randomisation hinders a definite conclusion, and only a minority of the patients completed the 5-year follow-up. The 5-year recurrence rate after PBI of uveal melanoma is expected to be <5%, but it is known that a large tumour size is a predictor of recurrence.[15 16] Recurrences in our series were mostly marginal, indicating an inadequate safety margin in these cases.

Damato recently published preliminary results of TSR with adjuvant ruthenium-106 brachytherapy in 112 cases.[17] Our results appear to be comparable with his series with regard to eye preservation and local tumour control. His preference for adjuvant irradiation is delivered via a 25 mm ruthenium plaque, adding a wider safety margin, which potentially lowers the recurrence risk. With a favourable conservation of a minimum VA of 6/60 in 58%, this approach continues to be an alternative procedure to neoadjuvant PBI. However, it implies open surgery at an untreated tumour with a potential risk of iatrogenic tumour cell dissemination. No orbital/episcleral recurrence was noted in our present series after neoadjuvant irradiation in contrast to adjuvant brachytherapy series, where extraocular recurrence was reported in 8 from 18 recurrences[8] and 16 from 57 recurrences,[6] respectively. When applying preoperative 'neoadjuvant' irradiation to the tumour, the surgical clearance margin can be placed very close to the tumour, hence minimising the surgical coloboma to a minimum that only needs to be insignificantly larger than the tumour itself. This allows surgical excision of tumours with larger tumour diameter, the limiting factor being 'only' the risk for surgically induced postoperative bleeding, and not necessarily the fear of inducing tumour recurrences due to incomplete tumour resection. The reduced surgical safety margin in our series is also the reason for the relatively large percentage of some postoperative tumour residua (66%), which, however, had no dismal consequences and which regressed during the longer postoperative follow-up.

With MRI-based PBI, a highly conformal homogeneous irradiation was realised, minimising radiation exposure to the remaining ocular structures.[9] We did continue to remove irradiated tumours with a thickness of >6–7 mm, and we see this concept to be supported by our data with regard to the higher rate of neovascular glaucoma[5] and visual loss[2 5] expected after irradiation alone. In our series, neovascular glaucoma was managed with retinal photocoagulation and antiglaucomatous medication. Using lid retractors, irradiation of the eyelid could be avoided in all but two cases.

We performed primary PBI followed by scheduled TSR in a prospective protocol in patients in whom we anticipated radiogenic and tumour regression complications due to the initial large tumour load ('toxic tumour syndrome').[18] Konstantinidis *et al* reported delayed TSR in 12 patients with a history of PBI after the development of exudative retinal detachment with or without neovascular glaucoma and published the results most recently. Surgery was done at a mean of 17.4 months after PBI, achieving flat retina in 10 patients.[18]

Local recurrence in our previous series was associated with largest basal tumour diameter, retinal detachment and older age at diagnosis, and resulted in a greater risk for metastasis.[8] We did not find this to be paralleled in the present series probably because the event of local recurrence was too rare compared with the total number of patients. We did not obtain cytogenetic information in both series, and our 5-year rate of metastasis of 40% is well consistent with the present literature in patients with comparably sized tumours.[19]

Apart from irradiation issues, the surgical technique of TSR is very similar to the earlier approaches.[20] Some modifications have been proposed to ease the procedure and reduce complications,[17] but still, the surgery appears to be rather challenging, requiring some experience in eye wall preparation. Vitreoretinal instrumentation and techniques in contrast have evolved substantially and gained popularity during the past years. Some surgeons might prefer tumour endoresection for removal of choroidal melanoma, but we think that TSR continues to be a reasonable treatment option in tumours that invade the ciliary body, and cannot be visualised completely even with modern ultra-wide field viewing systems during vitrectomy. Systemic complications related to the intraoperative lowering of the arterial blood pressure during TSR (to about 40–50 mm Hg mean pressure) were not evident in our present series.

We were able to reduce local tumour recurrence to a minimum with the approach described herein. We are well aware of the higher effort and costs associated with this approach, but believe that it is a valuable option since local tumour recurrence is at present the only tumour-related (and possibly metastasis-related) parameter that we are able to influence with our treatment. There is a definite need for a comprehensive patient counselling with regard to the expected functional outcome and possible complications of the procedure.

Additional vitreoretinal surgery was frequently done in our series, mostly in cases with intraocular haemorrhage (with or without retinal detachment). The incidence of significant intraoperative haemorrhage might be lowered by coagulation of the posterior ciliary arteries in the quadrant of the tumour as has been proposed by Damato (personal communication), a technique that was not routinely applied to all cases in this series.[17]

Whether our sequential radiosurgical approach is advisable as a primary procedure with tumour resection performed in a fixed time schedule after irradiation as described herein or as a salvage procedure in the 'toxic tumour syndrome' as being proposed by Konstantinidis *et al*[18] is still a matter of open debate.

In conclusion, we were able to demonstrate that, in our cohort of consecutively treated patients, introduction of neoadjuvant PBI was an effective option to further reduce the incidence of tumour recurrence in TSR of uveal melanoma.

Author affiliations
[1]Department of Ophthalmology, Charité Universitätsmedizin Berlin, Campus Benjamin Franklin, Berlin, Germany
[2]Department of Ophthalmology, DRK Kliniken Westend, Berlin, Germany
[3]Berlinprotonen at the Helmholtz-Zentrum Berlin, Lise-Meitner-Campus, Berlin, Germany
[4]Department of Radiotherapy and Radiation Oncology, Charité Universitätsmedizin Berlin, Campus Benjamin Franklin, Berlin, Germany
[5]Department of Radiotherapy, HELIOS Klinikum Emil von Behring, Berlin, Germany
[6]Department of Ophthalmology, Städtisches Klinikum, Dessau, Germany
[7]Augenärzte Kurfürstendamm 102, Berlin, Germany
[8]Department of Ophthalmology, Medical University Innsbruck, Innsbruck, Austria

Contributors GDW, DC and NEB have substantially contributed in the conception or design of the work, or the acquisition, analysis or interpretation of data and in reporting of the work described in the article, as well as drafting the work/revising it critically for important intellectual content and approved the version submitted. LM, LK and MHF have substantially contributed in the conception or design of the work and in reporting of the work described in the article, as well as drafting the work/revising it critically for important intellectual content and approved the version submitted.

Competing interests None.

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